PROJECT AND TEAM INFORMATION

Project Title

SmartCare: Next-Gen Digital Healthcare

Student / Team Information

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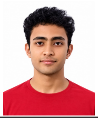
Team member 3	*Rawat, Sarthak Singh – 2029011* *2510350026@geu.ac.in* 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Healthcare institutions manage patient information, appointments and medical records using different systems and data structures. This can make efficient organization, retrieval and controlled sharing of information difficult. SmartCare aims to develop a smart healthcare management prototype using C++ Object-Oriented Programming and Data Structures and Algorithms. The system will manage patients, doctors, appointments, medical records and emergency cases. PostgreSQL will be used as the primary database for structured and persistent storage, while Firebase will be used for simulated cloud datasets and synchronization testing. The project will demonstrate efficient data handling using simulated healthcare data.

State of the Art / Current solution (1 pt)

Current healthcare management systems commonly use databases or hospital information systems for patient registration, appointments, medical records and billing. These systems provide storage and retrieval, but a student-level implementation may not clearly demonstrate the practical role of data structures in healthcare operations. SmartCare combines healthcare record management with Linked Lists, Queues, Priority Queues, Stacks and Hash Tables. PostgreSQL will act as the primary SQL database, while Firebase will provide simulated cloud datasets for testing cloud-based data access and synchronization.

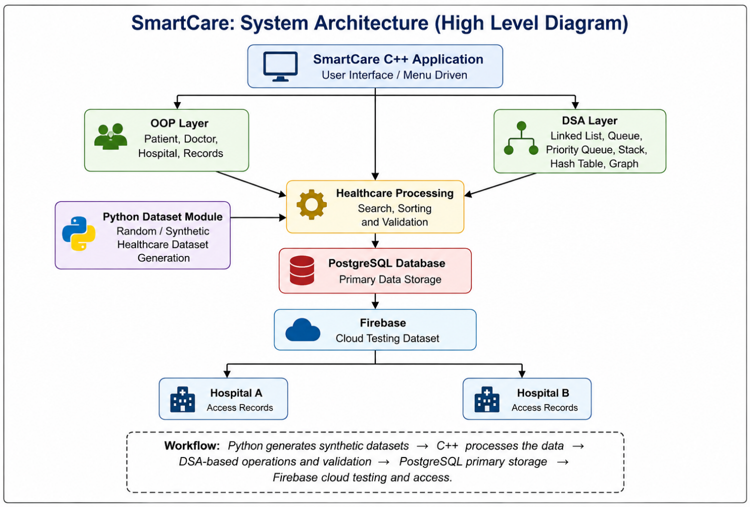
Project Goals and Milestones (2 pts)

1. *Design OOP classes for Patient, Doctor, Hospital, Medical Record and Appointment.*
2. *Use a Linked List for patient records and a Queue for appointments.*
3. *Use a Priority Queue/Min-Heap to prioritize emergency cases.*
4. *Use a Stack to maintain recent system actions.*
5. *Use a Hash Table for fast patient searching and record mapping.*
6. *Design PostgreSQL database for primary storage.*
7. *Integrate Firebase cloud service for synchronization testing.*
8. *Test the system using simulated healthcare records.*

Project Approach (3 pts)

The system will be developed as a modular C++ application. OOP will model healthcare entities and their relationships. DSA will be implemented directly in C/C++ and applied to practical modules: Linked List for patient records, Queue for appointments, Priority Queue/Min-Heap for emergencies, Stack for action history and Hash Table for fast lookup and data mapping. PostgreSQL will be the primary SQL database for persistent storage. Firebase will be used as a cloud-based testing environment containing simulated datasets and synchronization experiments. The application will provide menu-driven operations for adding, searching, updating and displaying records.

System Architecture (High Level Diagram) (2 pts)



Project Outcome / Deliverables (1 pt)

The expected outcome is a working C++ prototype, for healthcare management. It will show how to manage patients and doctors handle appointments prioritize emergencies, store records and track actions. The project will clearly use object-oriented programming concepts. Apply important data structures and algorithms in real ways. An SQL database will be used to keep data safe and available over time. Cloud integration will be shown as a feature to support remote backups and secure access. Final results will include the full source code, database design, sample test data, system architecture diagrams, documentation and presentation.

Assumptions

The system assumes the use of simulated healthcare data with a unique SmartCare ID for each patient. Only authorized users can access or share records between hospitals. PostgreSQL will store the primary structured data, Firebase will support cloud-based testing/backup, and Python will generate synthetic datasets for testing. The system is designed as an academic prototype with limited data and is not intended for real clinical use.

References

1. *GeeksforGeeks* – `https://www.geeksforgeeks.org/`
2. *TutorialsPoint* – `https://www.tutorialspoint.com/cprogramming/index.html`
3. ***Blockchain-Based Electronic Health Records Management: A Comprehensive Review and Future Research Direction***
   Authors: *Abdullah Al Mamun, Sami Azam, Clementine Gritti*
   Year: *2022*
   Publisher: *IEEE*
   Type: *Journal Article*
4. *For testing datasets, simulated healthcare data will be created using the NumPy library in Python and various machine learning models will be used for tuning and testing.*